Prelab 3

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Ans. to the Q. No. 1(a)

RMS value of a square wave

For one period, T = 2 sec, from fig. 1, signal f(t) is defined as:

$$f(t) = \begin{cases} 1 & 0 \le t \le 1 \\ 0 & 1 \le t < 2 \end{cases}$$

The RMS value is given by:

$$RMS = \sqrt{\frac{1}{T}\int_0^T (f(t))^2\,dt}$$

$$= \sqrt{\frac{1}{2}\left[\int_0^1 (1)^2\,dt + \int_1^2 (0)^2\,dt\right]}$$

$$= \sqrt{\frac{1}{2}(1+0)}$$

$$\therefore RMS\ value = \frac{1}{\sqrt{2}} \approx 0.707$$

Ans. to the Q. No. 1(b)

RMS value of a sawtooth wave

For one period, T = 1 sec, from fig. 2, signal f(t) is defined as:

$$f(t) = t \quad \text{for } -0.5 \le t < 0.5$$

The RMS value is given by:

$$RMS = \sqrt{\frac{1}{T}\int_0^T (f(t))^2\,dt}$$

$$= \sqrt{\frac{1}{1}\left[\int_{-0.5}^{0.5} t^2\,dt\right]}$$

$$= \sqrt{\left.\frac{t^3}{3}\right|_{-0.5}^{0.5}}$$

$$= \sqrt{\left(\frac{0.5^3}{3}\right) - \left(\frac{-0.5^3}{3}\right)}$$

$$= \sqrt{\frac{0.125}{3} + \frac{0.125}{3}}$$

$$= \sqrt{\frac{0.25}{3}}$$

$$= \frac{1}{\sqrt{12}}$$

$$= \frac{1}{2\sqrt{3}}$$

$$\therefore RMS\ value = \frac{1}{2\sqrt{3}} \approx 0.289$$

Ans. to the Q. No. 1(c)

RMS value of a sine wave

For one period, T = 4 ms, from fig. 3, signal f(t) is defined as:

$$f(t) = \underbrace{2.4}_{B}\sin\left(\underbrace{\frac{\pi}{2}}_{w}t\right) \quad \text{for } 0 \le t \le 4,$$ where peak amplitude (B) = 2.4 (maximum vertical displacement from 0)

period (T) = 4 ms (signal completes one full cycle from 0 ms to 4 ms)

angular frequency $(\omega) = \frac{2\pi}{T} = \frac{2\pi}{4} = \frac{\pi}{2}$ rad/ms

The RMS value is given by:

$$RMS = \sqrt{\frac{1}{T}\int_0^T (f(t))^2\,dt}$$

$$= \sqrt{\frac{1}{4}\int_0^4 (2.4\sin(\omega t))^2\,dt}$$

$$= \sqrt{\frac{2.4^2}{4}\int_0^4 \sin^2(\omega t)\,dt}$$

Now, $\int_0^4 \sin^2(\omega t)\,dt = \int_0^4 \frac{1-\cos(2\omega t)}{2}\,dt$; [We know trig identity: $\cos(2\theta) = \cos^2\theta - \sin^2\theta$

$\Rightarrow \sin^2\theta = \cos^2\theta - \cos(2\theta)$; $[\sin^2\theta + \cos^2\theta = 1]$

$\Rightarrow \sin^2\theta = 1 - \sin^2\theta - \cos(2\theta)$

$\Rightarrow 2\sin^2\theta = 1 - \cos(2\theta)$

$\therefore \sin^2\theta = \frac{1-\cos(2\theta)}{2}$]

$$= \frac{1}{2}\int_0^4 (1 - \cos(2\omega t))\,dt$$

$$= \frac{1}{2}\left[\int_0^4 dt - \int_0^4 \cos(2\omega t)\,dt\right]$$

$$= \frac{1}{2}\left[t\Big|_0^4 - \left.\frac{\sin(2\omega t)}{2\omega}\right|_0^4\right]$$

$[\omega = \frac{\pi}{2}]$

$$= \frac{1}{2}\left[[4-0] - \left[\frac{\sin((2)(\frac{\pi}{2})(4))}{2(\frac{\pi}{2})} - \frac{\sin((2)(\frac{\pi}{2})(0))}{2(\frac{\pi}{2})}\right]\right]$$

$$= \frac{1}{2}\left[4 - (\sin(4\pi) - \sin(0))\right]$$

$$= \frac{1}{2}\left[4 - (0-0)\right]$$

$$= \frac{1}{2}(4)$$

$$= 2$$

$$\therefore RMS = \sqrt{\frac{2.4^2}{4}(2)}$$

$$= \sqrt{\frac{2.4^2}{2}}$$

$$= \frac{2.4}{\sqrt{2}}$$

$$\therefore RMS\ value = \frac{2.4}{\sqrt{2}} \approx 1.697$$

Ans. to the Q. No. 2(a)

The frequency at which the magnitude (dB) drops by 3 dB from its max passband value.

From fig. 4 top figure, the passband gain is 0 dB.

Locating −3 dB on the magnitude curve corresponds to a frequency of $10^{-0.5}$ Hz.

$\therefore$ Cutoff frequency $10^{-0.5} \approx 0.316$ Hz.

Ans. to the Q. No. 2(b)

Based on the amplitude-frequency plot (fig. 5),

The filter has a max gain of 1 at low frequencies.

On a linear amplitude scale, the cutoff occurs when the amplitude is $1 \times \frac{1}{\sqrt{2}} \approx 0.707$.

In fig. 5, y-value 0.7 in the vertical Amplitude axis is found when frequency (x-axis) is around 1.5×10^5 Hz.

$\therefore$ Cutoff frequency $= 1.5 \times 10^5$ Hz

Ans. to the Q. No. 2(c)

In figure 6,

The filter has a max gain of 1 at low frequencies.

On a linear amplitude scale, the cutoff occurs when the amplitude is $1 \times \frac{1}{\sqrt{2}} \approx 0.707$.

In fig. 6, y-value 0.7 in the vertical Amplitude axis is found when frequency (x-axis) is around 1 (no units because x-axis is dimensionless frequency)

$\therefore$ Cutoff frequency = 1 (normalized).